Experiment – 4

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Title

Exploration vs. Exploitation Analysis using Multi-Armed Bandit Algorithms

AIM

To implement the **Epsilon-Greedy Multi-Armed Bandit Algorithm** and analyze the trade-off between exploration and exploitation in Reinforcement Learning.

PROCEDURE

1. Import NumPy and Matplotlib.
2. Define multiple bandit arms with different reward probabilities.
3. Initialize estimated rewards and arm selection counts.
4. Set the exploration probability ε.
5. Select a random arm during exploration.
6. Select the arm with the highest estimated reward during exploitation.
7. Generate rewards based on the selected arm.
8. Update the estimated reward of the selected arm.
9. Repeat the process for multiple rounds.
10. Display the estimated rewards and arm selection counts.
11. Visualize the number of times each arm was selected.

4. PYTHON SOURCE CODE

```
import numpy as np

import matplotlib.pyplot as plt


np.random.seed(42)


num_arms = 5

num_rounds = 1000

epsilon = 0.1
```

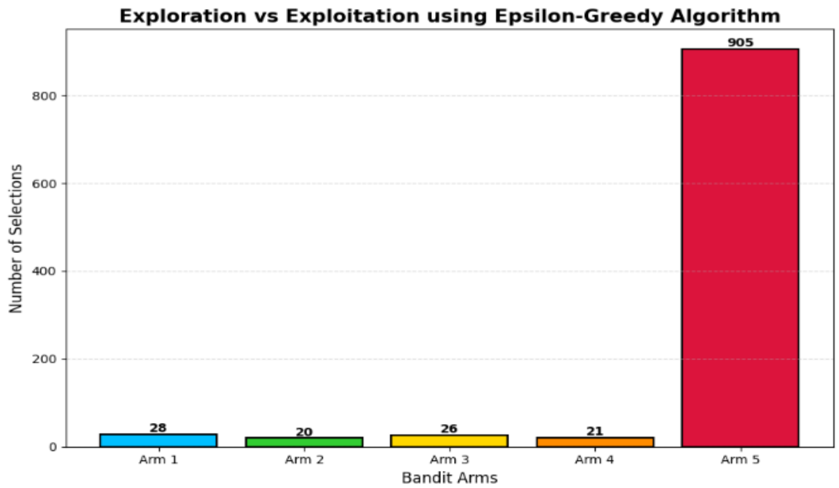
```
true_rewards = [0.2, 0.5, 0.7, 0.4, 0.9]

estimated_rewards = np.zeros(num_arms)
arm_counts = np.zeros(num_arms)

reward_history = []

for _ in range(num_rounds):

    if np.random.random() < epsilon:
        arm = np.random.randint(num_arms)
    else:
        arm = np.argmax(estimated_rewards)

    reward = 1 if np.random.random() < true_rewards[arm] else 0

    reward_history.append(reward)

    arm_counts[arm] += 1

    estimated_rewards[arm] += (
        reward - estimated_rewards[arm]
    ) / arm_counts[arm]

print("Estimated Reward for Each Arm")
print()

for i in range(num_arms):
    print(f"Arm {i + 1}: {estimated_rewards[i]:.3f}")

print("\nNumber of Times Each Arm Was Selected")
print()
```

```
for i in range(num_arms):
    print(f"Arm {i + 1}: {int(arm_counts[i])}")

print("\nTotal Reward:", sum(reward_history))

arms = ["Arm 1", "Arm 2", "Arm 3", "Arm 4", "Arm 5"]

plt.figure(figsize=(10, 6))

bars = plt.bar(
    arms,
    arm_counts,
    color="steelblue",
    edgecolor="black",
    linewidth=1.2
)

plt.title(
    "Exploration vs Exploitation using Epsilon-Greedy Algorithm",
    fontsize=15,
    fontweight="bold"
)

plt.xlabel("Bandit Arms", fontsize=12)
plt.ylabel("Number of Selections", fontsize=12)

plt.grid(
    axis="y",
    linestyle="--",
    alpha=0.5
)
```

```
for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width() / 2,
        height + 5,
        f"{int(height)}",
        ha="center",
        fontsize=10,
        fontweight="bold"
    )

plt.tight_layout()
plt.show()
```

OUTPUT

```
... Estimated Rewards
    Arm 1: 0.143
    Arm 2: 0.400
    Arm 3: 0.769
    Arm 4: 0.429
    Arm 5: 0.894

    Arm Selection Count
    Arm 1: 28
    Arm 2: 20
    Arm 3: 26
    Arm 4: 21
    Arm 5: 905

    Total Reward: 850
```



5. VISUALIZATION

The bar graph shows the **number of times each arm was selected** during the experiment. The arm with the highest reward probability is expected to be selected more frequently as the algorithm learns which action provides better rewards.

6. SUMMARY TABLE

Parameter	Value
Number of Arms	5
Number of Rounds	1000
Exploration Probability (ε)	0.1
Algorithm	Epsilon-Greedy
Highest Reward Probability	0.9
Visualization	Arm Selection Bar Graph

7. RESULT

The Epsilon-Greedy Multi-Armed Bandit algorithm was successfully implemented. The agent performed both exploration and exploitation and gradually selected the more rewarding arms more frequently.